AI and Ethics: Agriculture

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Did you know that agribusinesses who sell farmers seeds and chemicals increasingly trade in big data and AI? Not unlike Facebook, Google and Amazon, companies such as Bayer/Monsanto use data collected via smart tractors with machine learning to derive insights on farming that they sell back to farmers. In this talk, Bronson will use empirical examples to develop a closely observed account of farm data collection and use with AI in the agri-food sector. Bronson will draw attention to agriculture as a high impact use case by walking through some of the potential ethical and social justice implications of AI in this sector. Ultimately, the talk will explore what could happen to food sovereignty and environmental justice if we do not anticipate the ethics of agricultural AI.

Lecture Transcript

0:01 Hi, my name is Dr. Kelly Bronson. I am going to be telling you today about AI in the context of agriculture and some of the affordances, but also potential limitations of these emergent technologies. My background was originally in plant science and biology. And after being on the lab bench, I moved to the sociology of technology because I became interested in the public face of biotechnologies at the time. My upper degrees are in the social sciences. And I'm now a Canada Research Chair in Science and Society at the University of Ottawa, in Canada's capital city.

0:50 So I'm going to be talking, as I said, about AI ethics, but in the context of agriculture. So many people believe that we're in the midst of a digital revolution, in the context of food production, a revolution in food production led by innovations in digital technologies. So for example, tractors that are embedded with hundreds of sensors that passively collect data from farm fields, or on the farmer themself from the moment they open the cab door. And then those data collected from these pieces of precision equipment, it's called, are aggregated or put together with other kinds of data, for example, from unmanned aeronautical vehicles or drones, with satellite data, and they're made into big datasets, transferred to the cloud based infrastructure and then aggregated into big datasets by corporations like John Deere, the manufacturer of the big green tractor. And John Deere isn't alone. Other corporations are playing in this space.

2:08 And then these data sets are mined by machine learning to deliver insights on farming. And then corporations like John Deere are now selling those insights or that information to farmers. Farmers can access the information from a tablet or from a smartphone. And so this is what broadly together people are calling “digital agriculture”, an umbrella term for this suite of digital tools, including AI, perhaps at the center of which is AI. But people also use the term “Smart Farming”, even “Agriculture 4.0”. I'm going to today be using the shorthand AI in agriculture to include a lot of these digital tools or to capture that. One of the big hopes for AI in the context of agriculture is that, especially automation, will help farmers but also the whole sector in terms of supply management fill, what is thought to be a kind of perennial labor supply problem, especially in the soft fruit sector, where there are labor needs that are intermittent and and it's difficult to fill those labor needs.

3:26 Currently in Canada, as in other jurisdictions, we fill those labor needs with temporary workers. And they're mostly migrant laborers. And migrant laborers, you know, provide a solution to this labor, so-called labor problem in horticulture. But they themselves present a host of legal but also ethical challenges in how these members of the food system community are treated. And in fact, the labor problem is exacerbated by global issues or events. And one example and I grabbed a news headline on the bottom here of the slide was the Coronavirus pandemic. Because of border, a lack of border porosity, migrant workers faced problems getting home. And then in fact, many of them were subject to outbreaks that happened in, for example, food processing plants and other kinds of places where these workers find themselves.

4:35 So people are hopeful that automation will solve the labor supply problem, but also they're hopeful about other promises coming from AI in the context of agriculture. And here I just have a drop list of some of those promises. One of the promises, which, of course we see elsewhere, is the promise of economic development, especially in rural areas. So in Canada, as in other jurisdictions, rural employment is there, many rural communities are depressed economically. And so there's hope that with digital skilling and new kinds of jobs perhaps there's an opportunity in rural areas, especially with AI in agriculture. There's promises around improved animal welfare. So one of the domains where we see AI applications already being implemented and adopted is in milking in dairy farms, especially in New Zealand, Australia, but also in Canada. And here, those tools are being adopted for efficiency, including labor efficiency, but also improved animal welfare. So this is animals who bring themselves in basically guided by digital tools to milk. We also see increases in traceability, which, of course, is the benefit in terms of potential sustainability or animal welfare. Communication to consumers who are willing, for example, to pay more for food that's been grown or harvested under particularly humane or sustainable conditions. We also see a promise of increased productivity. So the idea is that with those insights that are data driven, or driven by machine learning algorithm mining a big data set, that farmers will be able to increase their yields and their productivity, which will then lead to a profitability gain and a competitive advantage for individual farmers.

6:44 And then if we scale up that opportunity around productivity, there's a promise that AI in agriculture can help deliver more societal wide or global wide food system benefits, like global food security. There's last sustainability promise around AI and agriculture, that just like in targeted medicine, for example, right, where you have a data driven recommendation for a drug treatment, the targeted recommendations for farmers, sometimes actually themselves called prescriptions by, for example, the company, Bayer/Monsanto, in their delivery of these AI insights. These prescriptions are prescriptions, not just about when to seed or when to harvest, but also what inputs to apply, including chemical inputs, and how much and where in the field. And just like with targeted medicine, the hope for AI in agriculture is that these recommendations will be so precise, and so perfected from what a human might do that they're going to lead to a more judicious use of harmful or scarce inputs, everything from chemicals, which we know are harmful to fossil fuels. And then at scale, right, especially in relation to fossil fuel consumption, if this can scale, this promise, the sustainability promise then we might see societal wide or even global environmental gains, like for example, climate change mitigation.

8:29 I just want to spend a little bit of time dealing with number six here, this idea of the sustainability promise of AI and agriculture. Because for people not familiar with the agriculture or the agri food sector, it's now well established that what we call conventional or industrial agriculture, and I have an image here of an industrial scale farm, right farming, mostly one thing or monoculture intensively, using inputs like chemicals, that that kind of farming has delivered benefits, especially in the form of increased productivity of certain kinds of commodities. And it's delivered benefits in terms of economic gain for certain types or scale of farmer and certainly for corporations supplying the inputs. But at the same time, it's well established now that intensification of agricultural practices via technologies, everything from mechanization to chemicals, has brought a host of ecological ills. And I just point you toward one report from a UN level organization "From Uniformity to Diversity" from 2016. So, and I emphasize the sustainability promise around AI and agriculture because this promise is widely shared. So this quotation is from the Head of Digital Agriculture for Bayer/Monsanto, who says that if we want to increase productivity, while safeguarding our food supply in the long term, then digitalization can help us do this right. It will help us deploy resources efficiently and sustainably, enabling farmers to get the best out of their fields with minimal environmental impact.

10:27 We also see this promise throughout the academic literature. So this was a very this quotation is from a very, very early and influential paper that was published in the journal, *Precision Agriculture*, where these scholars described AI and agriculture as leading to a paradigm shift, they thought the change will be so substantial in the substitution, as they say, of environmental information for physical inputs. And here, they really mean chemicals, that it's going to be paradigmatic. This is going to be a paradigm shift. Okay? So amidst this reality of agriculturally induced environmental harm, right, which I would say, is a consequence of environmental change as much as it contributes to it right? Farmers are made vulnerable by events like global climate change, and the ensuing weather volatility. There's a lot of hope around precision agriculture to help us meet these sustainability or environmental challenges. And this is just a snapshot. And this is from the USDA, of investment, right that's attached to this hope for some of the promises around AI in agriculture. So attached to this hope is a lot of investment not on the not just on the part of industry. So John Deere, for example, has invested hundreds of millions of dollars in transitioning itself from a machinery company to a data strategist. And we see very similar investment strategies among the other major input supply agricultural corporations, again, like Bayer/Monsanto, who have pivoted from investing in research and development among else in seeds and chemicals, for example, to data analysis, to tools for data collection, but also analysis and strategizing.

12:28 So we see investment from the private sector, but also from the public sector. So again, here's from the US Department of Agriculture, and then we also see investment on the part of individual farmers. Okay? So I don't dismiss this hope, for AI in agriculture, the hope that these new innovations in computational analysis and data collection, big data and machine learning will help us meet some of the challenges in our food system. But, you know, history has shown us and I'm a sociologist of technology, I'm not an historian, but part of my job entails taking the wide view. And history has shown us that innovations bring wins and losses. There are never wins across the board. And history has also shown us what happens when we fail to anticipate and mitigate the harms, from what is always inevitably social change that's led by technical change. So in that regard, my job as a social scientist is to try to help us think through and anticipate some of the otherwise unintended consequences, social justice, ethical consequences coming from new technologies.

13:50 So in the context of AI in agriculture, what might those consequences be? Well, here is a drop list. And I'm going to go into a fair amount of detail in all of these. So I'm calling it on the slide risks, which is, of course, always probabilistic, right? So risks as in potential harms from AI in agriculture. So here's a drop list of some of the ethical, social justice, as they also relate to environmental change, issues arising from AI in agriculture. One of the risks is around this technological displacement. So while I said, there's great hope, that automation, especially automation in the form of robotics but also AI, can help us meet labor supply challenges in the food system. That of course also means that certain people will be out of jobs. And one thing I wanted to point to in a discussion about potential technological unemployment or displacement is that this won't happen evenly. So while some agricultural laborers, right, like temporary foreign workers may find themselves out of work, it seems to me from my research, there is already an increasingly a need for skilled workers in data science, in agronomy or agricultural science, and especially in cybersecurity as it relates to agriculture. So the risks, the potential risks of technological unemployment will not be filled evenly. And in fact, to me, this is a social justice issue, because the displacement will map onto existing socio economic divides, right? Where the lowest skilled workers are likely to be displaced. And in fact, it's beyond I think, a social justice issue. It's a strategic or democratic issue, because we know that technological displacement from our experience in other sectors, like automotives for example, can lead to whole social groups feeling disenfranchised, and that itself can lead to social unrest, instability, right and a polarized political discourse, like we're experiencing, frankly, right now in the US and in Canada.

16:34 So that's one potential risk from AI in agriculture–is this risk of technological unemployment, and especially in the unevenness as it maps on to historic or existing disadvantage, we might call it or marginalization. Okay. So now to the second risk, which I called the unequal benefits, or we might say that incommensurate benefits coming from AI, not just among social groups, but among private sector actors and technology users. And we've seen this before, this risk. One of the interesting things to me when when I talk about AI in agriculture, or big data in agriculture, is that there's lots of attention to risks coming from the uses or misuses of personal data collected or harvested from internet behavior, and the reproduction of disadvantage, inequity or even harms like systemic racism. And a notable book here in this, in that sense, in thinking about these things outside of the agricultural context is, of course Zuboff's tome, *The Age of Surveillance Capitalism,* where that concept surveillance capitalism has been widely applied to the uses or again misuses of data collected from persons right through internet use. But these ideas have not been given attention, not to the same degree in the context of agriculture.

18:19 So this is something that I tried to do in my work. Think about what we might call the political economy around the uses and misuses of agricultural data. And I have a book on this subject, which goes into great detail. And I have article link treatments. And there's a lot of detail that I absolutely cannot cover in today's module. But here is a table that might be a helpful summary to which you can refer. So you know, this on the left hand side, the first column tracks data uses and AI uses in the context of agriculture. The second column is who's benefiting, right, which is fundamentally the question that people ask using concepts like surveillance capitalism. And then what are some of the potential issues right in the uses or potential misuses of agricultural data? So I'm just gonna go into a tiny bit of detail on a few of these.

19:28 Okay, I will say at the start that, just like big tech, what we classically think of as big tech, right, Facebook, Google, Amazon. Agricultural businesses, like the maker of the big green tractor, like Bayer/Monsanto, a chemical seed company, they are really pivoting or have pivoted to data companies. Where they make money, actually, this is a distinction from social media or internet or platform based data collection, where these companies are making money, because farmers actually pay for the data driven insights, right, unlike how we use a platform like Facebook. But also farmers arguably pay with their data. So there's evidence that data themselves are being acidized. Right, they are a commodity in and of themselves, that the companies control, and that they can sell, for example, to third parties, or they can use for their own purposes.

20:35 And I'm just going to give one example. And that's price setting. So that's the third row down here. So, you know, arguably, the data sets that can be mined by algorithms for insights, that predicted analytics could be very helpful. And there's some evidence that it already is being used this way to input supply companies to help them predict right whole regions or areas of farming that are going to need particular chemicals. And the companies in agriculture are so few, they're oligopoly corporations, that they operate outside of traditional market forces. And we already know these companies set prices. So there's no reason or we have good reason to infer that predictive analytics will be used by the input supply companies to set prices. And you know, this graph is an illustration of potentially the farm level consequences of this kind of AI driven price setting behavior. The debt that's taken on by individual farmers, in places like Canada and the US and Australia has grown asymptotically for the last 75 years with the introduction of new technologies. And what is sometimes called a cost price squeeze on farmers, right, where they get very little money for their commodities on the international market and they pay quite a bit of money for farm inputs. They thereby end up squeezed where the net farm income in Canada after costing is below zero and has been for some time on average.

22:32 And, you know, there are food system wide consequences, not just consequences to individual farmers. So the consequences of debt are such that, you know, people are not entering the farming industry. It's an economic losing proposition. But also there are mental health issues among farm communities, among others, among other things. But there's also a food system wide issue and that is, money begets money. And power begets power. So if these corporations, I'm going to go back 2 slides, right? If you see under the beneficiary's side, that the AI driven insights, well, surely they'll help individual farmers, those who pay for the insights. A lot of these data uses and algorithmic uses are going to really help corporations in an unequal or incommensurate fashion, right to give them predictive insights, to be able to set prices, to be able to secure or lock in relationships with farmers for product development, right where we know that those who collect data have an advantage in the space of machine learning. And on and on and on.

23:54 And this kind of data advantage will lead to an economic advantage and arguably increasing concentration in an already incredibly concentrated sector. So Phil Howard is an economist who does beautiful visual statistics and has a well stocked website I encourage you to go to. And here's just a basic bar graph illustration of concentration in the sector, in chemicals, in seeds and farm equipment, you can see just a handful of corporations globally, supplying these inputs to all farmers, all over the world. And there's no reason to think that because of just the one example I've given, but as I said in my book, I detail many others, that the uses of data and analytics and AI won't just further this market concentration and power. And that actually has a host of consequences, not least of which importantly important to me. My background is not just in biology, my earliest beginnings but environmental biology. And one of the major issues is that concentration as my colleague Jennifer Clapp says, in this article, among transnational corporations, the ones that sell seeds and agrochemicals, encourages the whole food system, right to contribute to a decline, or encourages agricultural systems like what we sometimes call the industrial system. And this system contributes to declines in plant genetic diversity and increasing use in chemicals. Okay.

25:34 So what are the other issues, I'll come back to this slide of Safiya Noble's book. But one of the other issues that I listed was bias, algorithmic bias. And, you know, again, just like with this idea of surveillance capitalism and the political economy of internet data usage, we can see similar patterns of or people with internet data have pointed out biases or skews in the working of algorithms and the feeding datasets that are used with, for example, search algorithms. So you know, Safiya Noble's, exemplary take on Google search, and how, because of skewed data, right search reproduced, effectively, systemic racism. And so while not the same, we can actually see similar sorts of bias within the commercial algorithms in agriculture, AI in agriculture. And here's just one example. Even if one looks, and I have published on this, articles on this. Even if one looks at just the types of data that are collected into the commercial, data driven, or AI, what's called decision support platforms for farmers. I mentioned that John Deere had one at the beginning. Bayer/Monsanto has an offer for farmers. If we look at the crops, that these commercial algorithms work with the crop types, the data that are collected, they're only data relevant to a particular style of farming. They're only data on what we might call agronomic or commodity crops. These are crops like corn, and soy, and feedstock corn, not all corn. And these are crops grown on relatively large farms. Farms that really have to grow at that scale, because the commodity prices are so low for those kinds of crops. But they're farmers that have access to lots of land, typically, relative to other farmers. And they also have access to capital because they tend to be larger, with more need for expensive machinery, but also access to capital. So we might call them relatively powerful farmers among farmers. Now, I want to just make clear that all farmers have a difficult time today. But but you know that we can see a kind of bias in the algorithmic offerings by the private sector, toward a particular style of agriculture, the one that correlates with environmental harm, as Jennifer Clapp says, indicates, but also toward a particular user, one that at least in terms of access to debt is relatively more powerful. And so and so we see a similar, it's not the same, but a similar kind of skew and bias that we see in the uses of big data and AI in other sectors in agriculture.

29:05 One of the other issues that I had on my drop list is that actually figuring out some of the potential harms or misuses from agricultural AI and data is nearly impossible in my experience, because the data sets, the algorithms and even the scientists who might tell us something about how the data were collected or weighted or how the model was developed, they're all obfuscated by legal mechanisms, copyright law, intellectual property law and trade secrecy law. And of course, we have seen historically this kind of lack of transparency and accountability in other sectors. But that has been blown up a little bit, mostly by whistleblowers and exposé and things have been worked out right in the context of social media platforms, not completely, not resolved, but somewhat worked out in the court of public opinion. But we just haven't seen that with agriculture. And so as a social scientist for my book project, for example, when I wanted to really follow the data, so to speak, I found they were what Frank Pasquale, the legal scholar, calls "a pernicious blackbox", which makes it actually really difficult to perform independent analysis, either from a critical social science or social justice perspective, but also even a scientific validity perspective, right? How can we know if these algorithms do what they're supposed to do or they say they're going to do for farmers or for the food system? How can we know if they work if we can't ever figure out or validate how they work? Okay.

30:58 So that's one issue, I think, on its own, which is this lack of transparency. And it actually relates to another issue, which and I put them together on the drop list, which is that there's a lack of regulation, legislation, regulation, or even governance more broadly, around agricultural AI and data usage. So for example, in Canada, we have legislation pertaining to the uses of personal data privacy legislation, none of it covers farm level data, in part because the agricultural AI and especially the data space is a bit unique, much of the data is environmental.

What do we do with that? There has been this sort of historic assumption that environmental data poses very little risk, vis-à-vis privacy, or other legal concerns. But there is all sorts of data collected by, for example, the sensing tractors that pertains to the person, but it's currently not covered under legislation. What we have currently is companies who themselves are volunteer, they have voluntary privacy policies, that they mostly detail in policies with technology users, whether it's a Terms of Use Agreement, or a licensing agreement. And I have for my book project, read those legal documents in detail. And I can say that, what my colleague in law at the University of Ottawa, Michael Geist, who's a copyright lawyer, he talks about this in the non agricultural sector, he calls it "virtue signaling". There's a kind of gesturing toward virtue and toward ethics by these companies, who in the terms of us talk about the importance of protecting farmer privacy, for example. John Deere says that they follow the NIST cybersecurity framework, and they're concerned about things like, you know, nation-state level cybersecurity, for example. And they have internal security strategies and data management policies. But it seems to me a lot of signaling at this point. And we need, and there's currently a gap in terms of legislation covering agricultural data and AI. Okay.

33:29 Another issue is a potential cybersecurity breach, or risk. In May of last year, CNN reported that John Deere machinery was stolen from farmers in the Ukraine by Russian actors. And interestingly, John Deere was able to brick or hit the kind of kill switch on these tractors. And that actually is because they have the centralization of the control that John Deere Corporation has centrally on all of its tractors through well, basically using copyright legislation. They have a digital lock that they placed on the tractors but interestingly, I think this story even though it in some ways, evidences perhaps the security benefit from the centralized control of tractors by John Deere Corporation, it raises I think, potential red flags for the future. Because John Deere, the data for example, collected from John Deere tractors gets sent to cloud based infrastructure. And John Deere controls these data, as I said, centrally in an Operations Control Center, which means that the corporation itself or that control center is a potential cybersecurity victim right? Where the whole food system is made vulnerable by this potential to hack tractors to use them for ransomware, to hack and shut down tractors by nefarious actors, for example, wanting to disrupt food systems and particular jurisdictions.

35:10 The other thing that this story I think indicates is not just the cybersecurity threat, but how we see in the context of agricultural AI issues in tension with one another. So a potential, do the digital lock that's on John Deere tractors, and this is something actually that has made the news has prevented farmers from tinkering with their own machinery. Citing that as a breach of this copyright, the interpretation of copyright, copyright law. So that's a social justice issue that's received some attention. And I pointed you in this slide to Ifixit.org, which has done some work in this area. But we see in this case, how that digital lock, John Deere company would probably say, helped, right? Prevent Russian actors stealing, right and purposing this, this stolen machinery? Okay.

36:-9 So I would say the last issue I wanted to talk about is this potential digital divide. Just as we've seen with other internet based technologies, we see it in AI in agriculture. For one, the average age of a farmer in places like Canada, and it's similar in the US, Australia, and other places in the Global North, is increasing every Agricultural Census. Right now it's 56. And we know that age is correlated to skill, digital skill level, and also comfort with digital tools. So that's just one potential, right? At the very least, the AI revolution in agriculture might put an increasing burden on farmers to develop novel skills, right at a time in their life, when they may not have an interest in or be, frankly capable of doing. There are other digital divide issues as well. There's an uneven and a lack of access in rural areas to broadband connectivity, which you can't have a fully connected smart farm, or even a fully functioning cloud based infrastructure to enable the AI decision support platforms if you don't have reliable internet. Okay.

37:25 So all this to say there are a host of potential ethical, or I might say social justice issues arising from AI in agriculture. So while the tools may promise benefits and present benefits, there are risks. And really it's going to be these things which catch us out, right. So for those who are really enthusiastic about the potential benefits to the food system, or to individual farmers from AI in agriculture, we need to think about these ethical and social justice issues. Because we know from history, they lead to potential controversy. They've led to controversy, sorry. And they lead to issues with technology adoption among else, right. There are controversial precedents in the food system, like the case with biotechnology or GMOs, right. Examples where potentially a technically sound technology just didn't resonate well with members of the public or certain stakeholders, and led to real disruption in technology adoption and legitimation crisis on the part of governments, maybe specifically regulatory agencies.

38:42 But the good news is we have a framework. We have a framework for anticipating the potential impacts, and including a diversity of stakeholder or rights holder needs. Now, as we develop these tools for AI and agriculture. And that rubric or framework is called Responsible Research and Innovation, which I have the principles here on the slide, right, which really at the center of this rubric is inclusivity. There's a need to anticipate not just the impacts, but also include a diversity of food systems stakeholders in helping to anticipate those impacts and a diversity of needs and concerns. Okay.

39:27 There are examples actually, where user interests are being included in the development of AI based or in this case, it's robotics technology, but it's happening very piecemeal. So I just give the example here of dot robotics in Saskatchewan, where users have been engaged. There was a shift in the design, specifically around the connectivity issues, and there were amendments made to the robot and amendments made to the licensing model based on user concerns, but this is totally ad hoc. And this is based at the level of a company. And at least in Canada, but this isn't true of every place. So in Australia, I would say there's more nation state level capacity, at the level of the EU, there is as well. But in certain jurisdictions, like in Canada and in the US, there's really not the institutional capacity currently for thinking about a responsible AI revolution in agriculture. Okay.

40:35 And I would say this intersects with the regulatory issue, because, you know, we have proposed legislation in Canada, and I know some legislation is about to drop in the US and in the UK, for example. So in Canada, we have a Bill C-27, that's going to cover or is proposed to cover, be an update in legislation covering personal data usage and also AI. But you know, so we have propositions at least for regulations that account for things like data privacy, that account for things like data ownership issue, which, of course, I've mentioned is important, that account for things like data bias, as it intersects with other kinds of societal harm or disadvantage. I've indicated that's an issue. And to some degree, that accounts for the lack of transparency and accountability on the part of especially private sector actors doing things with data collected from people. But the legislation doesn't really account for non personal data, as I've said many times, and to me, the more fundamental issue is that actually, there's a first order question that has not yet been asked, that is essential to realizing a responsible agricultural revolution vis-à-vis artificial intelligence, and that is, how do we ensure that AI is aligned, right is, is developed in alignment with a diversity of human values?

42:13 I might just say, human values, but that responsible innovation principle of inclusivity and diversity that's really important, it's really essential here. Because actually, you know, the biases we see often in the development of algorithms that are skewed and reproduce harm that comes about not out of malice on the part of engineers, but often because the engineering community is itself biased, right? And so in the context of agricultural AI, what I have found in my research is that you have scientists, well meaning scientists, but they all share a very insular definition of what counts as good food, what counts as good society in the future, and that view perhaps needs to be diversified.

43:00 And so I think what we need is actually a consultation on this question. Before we get to legislation that deals with privacy and ownership, and maybe even with transparency, we need a public discussion on what legislation or regulation should prioritize? What are the principles that should serve as the foundation for regulation? And this is really the first step right. We've got supranational level principles like the Montreal Declaration and the OECD Principles on AI. But what we need is a jurisdictional and sector specific consultation and principles or a scoping of regulation. And this hasn't been done with AI. And it certainly hasn't been done in the context of agriculture and AI. Thank you. Here are some sources, and they'll be available through my slides as well.